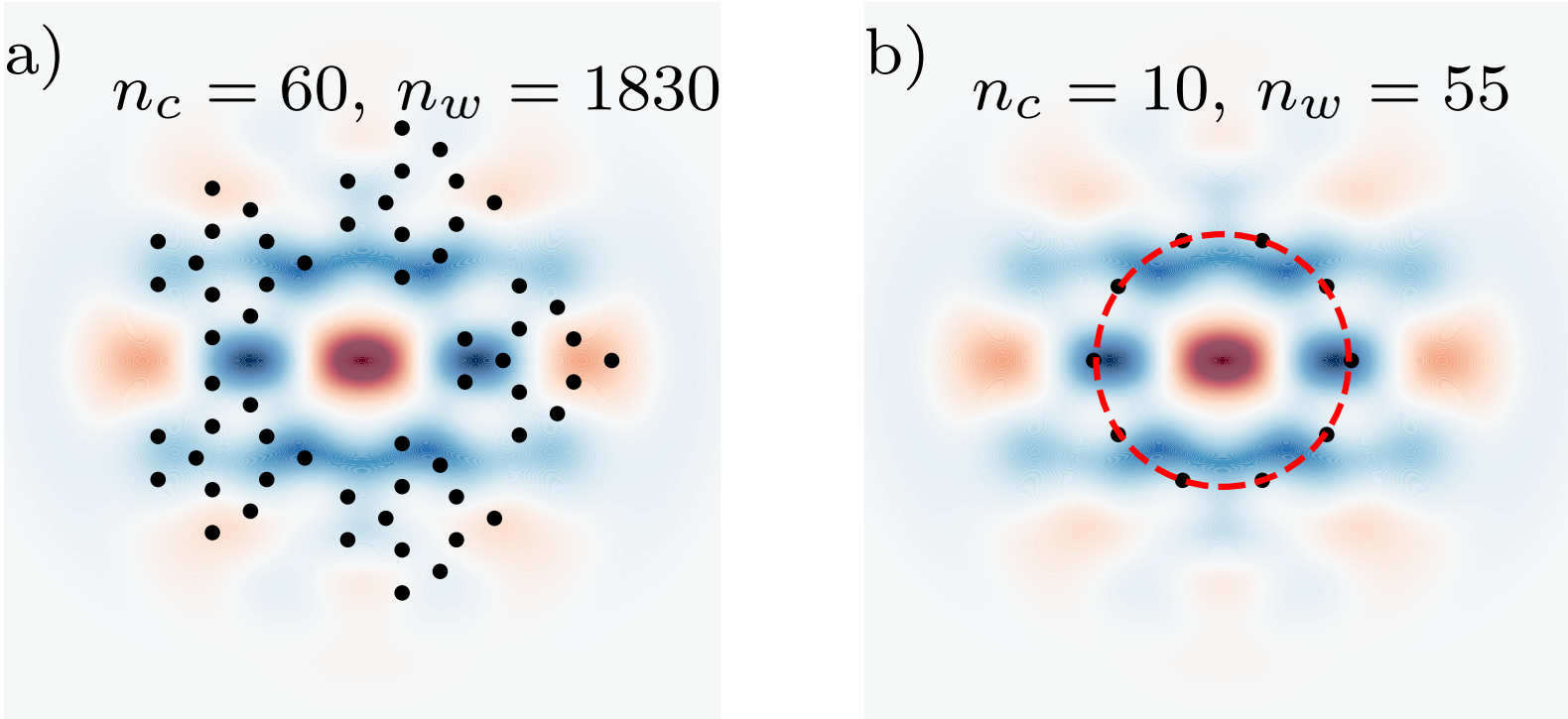

a)
$n_c = 60,\ n_w = 1830$
b)
$n_c = 10,\ n_w = 55$